Worksheet 9.1

Identifying and classifying minerals

Answers

Identifying minerals

1. Two common rocks are composed of the minerals shown in the images at right. Limestone contains calcite, and sandstone contains quartz.

Calcite

Quartz

The following table provides information about the properties of these minerals.

Mineral	Colour	Streak (colour of powdered mineral)	Hardness	Lustre
Calcite	Colourless, white	White	3	Dull to vitreous
Quartz	Colourless, purple, red, yellow	White	7	Vitreous

(a) Explain how calcite crystals can be distinguished from quartz crystals.

Quartz is much harder, so a Mohs' scale hardness test will show the difference.

(b) Pure quartz crystals are colourless. Explain the large variation in colour indicated in the table.

Coloured impurities can lead to significant colour changes.

Creating a classification key

2. Use the following table to create a written classification key for these minerals.

Mineral	Colour	Streak	Hardness	Lustre
Chalcopyrite	Yellow	Green, black	3.5–4	Metallic
Chlorite	Grey	White, green	2.2–5	Pearly
Galena	Grey	Grey	2.5	Metallic
Haematite	Grey	Red, brown	5–6	Metallic
Magnetite	Black	Black	5.5–6.5	Metallic
Pyrite	Yellow	Green	6–6.5	Metallic

Calcite photo: madllen / 123RF.com
Quartz photo: thomaseder / 123RF.com

Key:

1A. Mineral is grey ....**go to 2**
1B. Mineral is not grey ....**go to 4**
2A. Streak is grey.. **galena**
2B. Streak is not grey..**go to 3**
3A. Mineral has a metallic lustre ....**haematite**
3B. Mineral has a pearly lustre ....**chlorite**
4A. Mineral has a hardness less than 5.. **chalcopyrite**
4B. Mineral has a hardness of 5 or greater..**go to 5**
5A. Mineral is black...**magnetite**
5B. Mineral is yellow ....**pyrite**